This week continues the statistical foundations from Week 6 and moves toward a crucial question: **How do we design experiments and interpret evidence without allowing uncertainty, flawed reasoning, or our own expectations to distort the conclusion?**

The transcript covers six lectures, 37–42. I'll use the transcript as the backbone, distinguish the lecturer's explanations from methodological qualifications, and connect the ideas to practical scientific and AI/ML research.

1. Source and scope

Item	Details
Week	7 — Module 7
Lectures	37–42
Transcript quality	Mostly readable; some mathematical notation and speech-to-text passages need reconstruction
Main themes	Proportion measurement, box plots and outliers, propagation of errors, hypothesis testing, Type I and Type II errors, confirmation bias, controls, randomisation and blinding
External verification	Important statistical and experimental-design claims require qualification
Source limitation	Equations are sometimes spoken or transcribed imperfectly; where the intended equation is clear, I'll reconstruct it and state its assumptions

Lecture map

- **Lecture 37 — Examples of Proportion Measurement:** Sampling proportions, normal approximations, uncertainty and a one-sided hypothesis test using a survey of students.
- **Lecture 38 — Box and Whisker Plot:** Median, quartiles, interquartile range (IQR), whiskers and the responsible reporting of outliers.
- **Lecture 39 — Propagation of Errors, Part 01:** How uncertainty in measured quantities affects calculated quantities, including sums, differences, products and logarithms.
- **Lecture 40 — Propagation of Errors, Part 02:** Propagation through a function of one or more variables using derivatives and partial derivatives.
- **Lecture 41 — Issues in Hypothesis Testing, Part 01:** Testable predictions, logical limits of hypothesis testing, Type I and Type II errors, and confirmation bias.
- **Lecture 42 — Issues in Hypothesis Testing, Part 02:** Drug-testing example, null and alternative hypotheses, representative sampling, experimental and control groups, placebo effects, and single- versus double-blind designs.

2. Executive understanding

What you should understand after this week

1. **A measured proportion is an estimate, not the population truth.** Its uncertainty depends on the sample size and the underlying proportion.
2. **A normal approximation being reasonable does not mean the estimate is precise.** Approximation quality and uncertainty width are different questions.
3. **A box plot describes distribution, not just a central estimate.** It helps reveal spread, skew, and potential outliers.
4. **Outliers deserve investigation, not automatic deletion.** They can reflect mistakes, unusual conditions, or genuine scientific phenomena.
5. **Uncertainty propagates through calculations.** If you infer one quantity from other measured quantities, uncertainty in the inputs affects the output.
6. **A hypothesis is tested through its predictions.** Observing a predicted outcome does not logically prove the hypothesis that predicted it.
7. **A statistically non-significant result does not establish that there is no effect.** It may reflect low power, noisy measurements, or insufficient data.
8. **Confirmation bias can enter at every stage:** choosing a hypothesis, selecting participants or samples, collecting measurements, deciding what to exclude, and interpreting results.
9. **Control groups, randomisation and blinding solve different problems.** They help distinguish intervention effects from alternative explanations and reduce bias.
10. **A rigorous experiment is designed before the results are known.** Researchers should specify how data will be collected, analysed, and interpreted before they can be tempted to favour a desired conclusion.

The one big idea

Week 7 is about protecting the path from data to conclusion. Measurements are uncertain; calculated quantities inherit uncertainty; hypotheses can be tested only through evidence and assumptions; and researchers themselves can introduce bias. Good methodology therefore requires not just mathematical tools, but also experimental design, transparency, and disciplined interpretation.

3. Deep conceptual notes

Lecture 37 — Examples of Proportion Measurement

3.1 Why proportions need special treatment

[LECTURE] The lecturer contrasts measurements of continuous quantities—such as mass—with measurements of proportions, where each observation can be represented as 0 or 1.

Examples:

- Continuous measurement: mass of a sample, temperature, length.

- Binary measurement: a plant is tall or short; a test is positive or negative; a model's answer is correct or incorrect.

For a binary observation X_i,

$$X_i \in \{0, 1\}$$

The sample proportion is

$$\hat{p} = \frac{X}{n} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

where:

- n = number of observations
- X = number of observations classified as 1
- p = unknown population proportion
- $\hat{p}$ = observed sample proportion

If 33 out of 50 plants are tall,

$$\hat{p} = \frac{33}{50} = 0.66$$

The observed proportion is 66%. It does not follow that the population proportion is exactly 66%.

3.2 Sampling uncertainty in a proportion

If observations are independent Bernoulli trials with the same probability p,

$$E[\hat{p}] = p$$

and

$$\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

Therefore, the standard error of the sample proportion is

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Since p is unknown, we commonly estimate the standard error using $\hat{p}$:

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Intuition: Imagine repeatedly drawing independent samples of the same size from the same population. Each sample produces a slightly different proportion. The standard error describes the typical sampling variation of that estimator under the model.

As n increases, the standard error generally decreases at a rate proportional to $1/\sqrt{n}$. To halve the standard error, you typically need about four times as many independent observations.

3.3 Why "more than 25 observations" is not a universal rule

The lecture carries forward a rule of thumb from the preceding week: approximately 25 observations may be adequate for some normal approximations to sample means. It correctly emphasises that proportions need their own assessment.

For a binomial proportion, a common normal-approximation check is

$$np \gtrsim 10, \qquad n(1-p) \gtrsim 10$$

These are practical guidelines, not a universal theorem guaranteeing that every approximation will be accurate.

- If p is close to 0.5, the distribution is more symmetric.
- If p is near 0 or 1, it can be strongly skewed, and a larger sample may be necessary.
- For small samples or extreme proportions, Wilson or exact binomial confidence intervals can be preferable to the simple normal-based interval.

Critical distinction: The conditions for a good normal approximation are not the same as the conditions for a narrow confidence interval. A normal approximation can be adequate while the estimate is still too imprecise for the research question.

3.4 Worked example: 33 successes out of 50

The lecture revisits the earlier example:

$$n = 50, \qquad X = 33, \qquad \hat{p} = 0.66$$

Estimated standard error:

$$\widehat{SE} = \sqrt{\frac{0.66(0.34)}{50}} \approx 0.067$$

A rough 95% Wald interval is

$$0.66 \pm 1.96(0.067) \approx [0.529, 0.791]$$

This interval is wide: the data do not estimate the population proportion to within just two percentage points.

The lecture's calculation asks for the probability that a repeated-sample estimate falls within 0.02 of the true proportion. Using the normal approximation,

$$z = \frac{0.02}{0.067} \approx 0.30$$

and

$$P(|Z| \leq 0.30) \approx 0.236$$

So the probability is approximately **24%**, consistent with the lecture's rounded figure of about 23.6%.

The lesson is that 50 observations can estimate a proportion quite poorly when the desired margin of error is narrow.

3.5 Worked example: increasing the sample to 500

Now suppose $n = 500$ and $X = 330$, so $\hat{p} = 0.66$ again.

$$\widehat{SE} = \sqrt{\frac{0.66(0.34)}{500}} \approx 0.0212$$

The probability that the sample proportion falls within 0.02 of the true proportion, using the normal approximation and treating $p \approx 0.66$, is

$$z = \frac{0.02}{0.0212} \approx 0.94$$

Thus,

$$P(|Z| \leq 0.94) \approx 0.652$$

This is approximately **65%**, matching the lecture's result.

Even increasing the sample tenfold does not give 95% probability of being within a margin of just 0.02. This demonstrates why sample-size planning must begin with a desired precision or detectable effect, not a memorised minimum number of observations.

3.6 Worked hypothesis test: should internet access be blocked?

[LECTURE] An institute considers blocking internet access after midnight if more than 30% of students are using it for non-academic purposes and losing sleep.

A survey finds 57 such students out of 150:

$$\hat{p} = \frac{57}{150} = 0.38$$

The lecturer proposes a one-sided test. A more precise formulation is:

$$H_0 : p \leq 0.30 \qquad \text{versus} \qquad H_1 : p > 0.30$$

To calculate the largest tail probability allowed by the null, evaluate the test at its boundary, $p_0 = 0.30$.

$$SE_0 = \sqrt{\frac{0.30(0.70)}{150}} \approx 0.0374$$

$$z = \frac{0.38 - 0.30}{0.0374} \approx 2.14$$

The one-sided normal-approximation p-value is approximately 0.016.

At the 5% significance level, this is evidence against the null hypothesis. But this is **not enough by itself to justify the policy**. The decision also depends on whether the survey sampled students appropriately, whether responses were honest, whether “non-academic use” was clearly defined, and whether blocking access is an effective and proportionate response.

The lecturer's numerical illustration demonstrates the logic of a one-sided test. The policy decision is a separate problem requiring substantive evidence and consideration of consequences.

Lecture 38 — Box and Whisker Plot

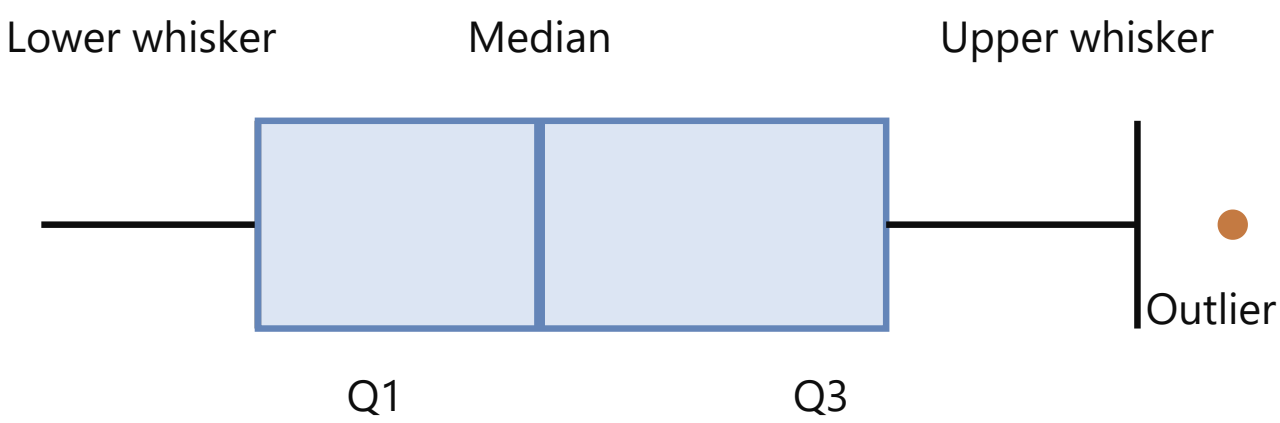
3.7 Why a mean and error bar are not always enough

A mean with an error bar communicates a central estimate and some measure of uncertainty or variability. But it may conceal important distributional features:

- Are observations tightly clustered or widely spread?
- Is the distribution symmetric or skewed?
- Are there multiple clusters?
- Are some observations unusually far from the rest?
- Do two groups have similar means but very different distributions?

A box plot provides a compact view of the distribution's quartiles and potential outliers.

Anatomy of a box plot



Schematic only. In a conventional Tukey box plot, whiskers usually extend to the most extreme observations within the 1.5 × IQR fences; they do not necessarily end at the dataset's absolute minimum and maximum.

3.8 The five-number summary and the IQR

A traditional box plot is built from:

1. Minimum or lower whisker endpoint, depending on the convention
2. First quartile Q_1
3. Median Q_2
4. Third quartile Q_3
5. Maximum or upper whisker endpoint, depending on the convention

The interquartile range is

$$IQR = Q_3 - Q_1$$

It measures the spread of the middle 50% of observations and is less sensitive to extreme values than the range.

How to interpret it:

- The median marks the centre of the ordered observations.
- The box spans the middle half of the distribution.
- A longer box indicates greater spread in that central half.
- Unequal distances on either side of the median can suggest asymmetry.
- Points beyond the whiskers are flagged for investigation.

3.9 The 1.5 × IQR rule

The conventional Tukey fences are

$$L = Q_1 - 1.5(IQR)$$

$$U = Q_3 + 1.5(IQR)$$

Observations below L or above U are commonly flagged as potential outliers.

A subtle but important distinction: **the fences are not the whiskers themselves**. The whiskers generally extend to the most extreme observed values still inside those fences. Points beyond them are plotted separately.

These are exploratory flags, not formal proof that a point is erroneous. The 1.5 × IQR rule is a convention for identifying unusual observations, not a universal law of nature.

3.10 What should a researcher do with an outlier?

The lecture's ethical point is valuable: an unexpected measurement must not simply disappear because it conflicts with expectations.

Possible causes include:

- Instrument malfunction or recording error
- Data-entry or unit-conversion mistake
- A different experimental condition
- A genuine but uncommon phenomenon
- A heavy-tailed or otherwise poorly modelled distribution

A responsible procedure is:

1. Preserve the original observation and audit trail.
2. Check measurement records, units, equipment and data-processing steps.
3. Apply exclusion criteria defined independently of whether the result supports the hypothesis.
4. Report exclusions and the reasons for them.
5. Where appropriate, show results with and without the observation, and discuss how sensitive the conclusion is to it.

Important qualification: A point being flagged as an outlier does not automatically justify excluding it from all later calculations. If there is a defensible measurement error, correction or exclusion may be justified. If it is a valid observation, robust methods or a model that accommodates the data may be more appropriate.

Transcript example: numerical inconsistency

The lecture's example dataset lists 14 values, but the transcript later says there are 13 and gives a sorted list that omits one occurrence of 16.3. The reported quartile calculations are also inconsistent with the listed data. Consequently, the general method above is reliable, but the lecture's numerical quartiles should not be memorised as a correct calculation from that dataset.

This is precisely the kind of situation in which a researcher should verify the raw observations before accepting a derived result.

Lectures 39–40 — Propagation of Errors

The central problem is straightforward:

> If I measure x and y, but the quantity I actually need is $z = f(x, y)$, how should uncertainty in the inputs affect uncertainty in the result?

This matters whenever a quantity is calculated indirectly rather than measured directly.

3.11 Propagation for a sum or difference

Suppose two measured quantities are reported as

$$x = \bar{x} \pm \delta x, \qquad y = \bar{y} \pm \delta y$$

For the sum,

$$z = x + y$$

the central estimate is

$$\bar{z} = \bar{x} + \bar{y}$$

If the uncertainties represent standard deviations of independent random errors, then

$$\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$$

The same rule applies to the difference $z = x - y$:

$$\bar{z} = \bar{x} - \bar{y}, \qquad \sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$$

Why doesn't the sign of the operation change the uncertainty formula? Because variance involves squared deviations. Under independence, the variances add whether the central values are added or subtracted.

Example: Suppose

$$x = 10.0 \pm 0.3, \qquad y = 7.0 \pm 0.4$$

where the uncertainties are independent standard uncertainties.

For $z = x - y$,

$$z = 3.0$$

and

$$\sigma_z = \sqrt{0.3^2 + 0.4^2} = 0.5$$

So the result is 3.0 ± 0.5, in the same uncertainty convention.

Notice that the absolute uncertainty, 0.5, is substantial relative to the result 3.0. Subtracting similar quantities can leave a small difference while retaining appreciable uncertainty. This is why cancellation can make *relative* uncertainty large.

3.12 Propagation for products and quotients

For

$$z = xy$$

take logarithms:

$$\ln z = \ln x + \ln y$$

For small relative uncertainties, linearisation gives

$$\frac{\delta z}{z} \approx \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

provided the input uncertainties are independent and the quantities are suitably away from zero.

For a quotient $z = x/y$, the same approximate relative-uncertainty formula applies:

$$\frac{\delta z}{z} \approx \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

Example: If x has a 2% relative uncertainty and y has a 3% relative uncertainty, then for their product the approximate relative uncertainty is

$$\sqrt{0.02^2 + 0.03^2} \approx 0.0361$$

or about 3.6%.

This is not the same as adding the percentages to obtain 5%. Adding absolute maximum error bounds and combining independent random standard uncertainties are different operations.

3.13 Propagation through a function of one variable

Suppose

$$z = f(x)$$

and x is measured near $\bar{x}$. For small deviations, a first-order Taylor approximation gives

$$\Delta z \approx f'(\bar{x})\Delta x$$

Thus, the approximate standard uncertainty is

$$\sigma_z \approx |f'(\bar{x})|\,\sigma_x$$

The derivative is a **sensitivity coefficient**: it tells you how much the output changes for a small change in the input.

- If $|f'(\bar{x})|$ is small, the output is locally insensitive to input error.
- If $|f'(\bar{x})|$ is large, input uncertainty can be amplified.
- If the function is strongly nonlinear across the uncertainty range, first-order propagation may be inaccurate.

Example: Let $z = x^2$ and suppose $x = 3.0 \pm 0.1$.

Then

$$z = 9.0, \qquad f'(x) = 2x$$

and

$$\sigma_z \approx |2(3.0)|(0.1) = 0.6$$

So the first-order approximation gives $z \approx 9.0 \pm 0.6$.

3.14 Propagation through a function of multiple variables

For

$$z = f(x, y)$$

the first-order perturbation is

$$\Delta z \approx \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y$$

If the input errors are independent,

$$\sigma_z^2 \approx \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2$$

All derivatives are evaluated at the measured central values.

If x and y are correlated, an additional covariance term is required:

$$\sigma_z^2 \approx \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\,\mathrm{Cov}(x, y)$$

This is an important precision upgrade to the lecture's derivation. The transcript suggests that positive and negative products of deviations cancel to zero with many observations. That cancellation is not guaranteed: it corresponds to assuming zero covariance or uncorrelated inputs.

The general law of propagation of uncertainty, including covariance terms and the first-order approximation, is documented by NIST. NIST +1

3.15 What “error” means here

The transcript uses *error*, *standard error* and *standard deviation* somewhat interchangeably. For rigorous research, keep these separate:

- **Measurement error:** Difference between a measured value and the quantity's true or reference value.
- **Standard deviation:** Spread of observations.
- **Standard error:** Estimated standard deviation of a statistic across repeated samples.
- **Standard uncertainty:** Quantified uncertainty associated with a measurement result.
- **Systematic error/bias:** A persistent deviation that does not necessarily average away with more repetitions.

The root-sum-of-squares rule is not a magical cure for every source of error. It combines appropriate uncertainty components under a model. Systematic effects, dependence, and model misspecification still require attention.

4. Lectures 41–42 — Issues in Hypothesis Testing

These are arguably the most important lectures of the week for developing research judgement.

4.1 A hypothesis is tested through its consequences

[LECTURE] The lecturer returns to the logical structure of scientific testing:

$$H \Rightarrow P$$

Here, H is a hypothesis and P is a prediction or observable consequence derived from it, together with any necessary background assumptions.

Two logical cases matter:

- If the hypothesis predicts P, but P is not observed under conditions where it should have been observed, the hypothesis may be challenged.
- If P is observed, that does not prove H, because other hypotheses may also predict P.

The second mistake is called **affirming the consequent**:

$$H \Rightarrow P, \quad P \quad \therefore H$$

That inference is invalid.

For example:

- Hypothesis A: A particular model architecture improves medical-image classification because it captures fine-grained spatial features.
- Prediction: The model should outperform a suitable baseline on a prespecified evaluation.
- Observation: The model does outperform the baseline.

This result supports the hypothesis only to the extent that competing explanations are addressed. The improvement might instead come from more parameters, extra training time, data leakage, hyperparameter tuning, or an implementation difference.

A crucial qualification: failed predictions

The lecturer argues that if a prediction is false, the hypothesis can be falsified by *modus tollens*:

$$H \Rightarrow P, \quad \neg P \quad \therefore \neg H$$

This is logically valid when the premise $H \Rightarrow P$ is genuinely established.

In real experiments, however, the prediction usually depends on more than one hypothesis:

$$H \wedge A \wedge M \wedge C \Rightarrow P$$

where, illustratively:

- H = main hypothesis
- A = auxiliary assumptions
- M = measurement model
- C = experimental conditions

If P fails, at least one part of this combined package may be wrong. The result is not always a clean refutation of H alone.

Research habit: When a prediction fails, inspect the main hypothesis, the implementation, the measurement process, and the auxiliary assumptions before deciding what the result refutes.

4.2 Scientific hypotheses should be testable

The lecture emphasises that a scientific hypothesis should generate predictions that can be examined experimentally or observationally.

A useful hypothesis should be sufficiently specific that researchers can state:

1. What observation would count as evidence for it?
2. What result would count against it?
3. What assumptions does that test depend on?
4. What alternative explanations remain?

Consider two statements:

- Vague: “This model understands medical images better.”
- Testable: “Under a prespecified evaluation on held-out cases, the proposed method will reduce false-positive findings relative to a specified baseline, without an unacceptable increase in false negatives.”

The second statement is not automatically a good hypothesis, but it is more operational. It identifies an outcome and a comparison that can be tested.

Why failed hypotheses can still be productive

The lecturer makes an important point: proposing a plausible, testable hypothesis that turns out to be wrong is not necessarily a scientific failure.

A well-designed negative result can:

- eliminate a proposed mechanism;
- reveal a hidden assumption;
- identify a dataset limitation;
- narrow the space of plausible explanations;

- prevent other researchers from repeating an unproductive line of investigation.

The standard of success is not whether the preferred hypothesis wins. It is whether the investigation provides trustworthy information.

4.3 Null and alternative hypotheses

The lecture introduces the conventional notation:

- H_0: null hypothesis
- H_1 or H_A: alternative hypothesis

For a drug study, a simplified formulation might be:

$$H_0 : \text{The treatment has no effect}$$

$$H_A : \text{The treatment has an effect}$$

In a real study, these statements must be made more precise. "No effect" could mean no difference in a mean, no difference in a probability, or no effect on a specified outcome under a specified model.

The null hypothesis is not necessarily the opposite of every conceivable alternative. A null such as $p = 0.30$ and an alternative such as $p > 0.30$ leave $p < 0.30$ outside the alternative being tested.

What a hypothesis test actually does

A statistical test asks how compatible the observed data are with a specified null model, using a chosen test statistic and its reference distribution.

The p-value is the probability, **assuming the null model and the test's assumptions**, of obtaining a test statistic at least as extreme as the observed one in the direction or directions defined by the test.

A p-value is not:

- the probability that H_0 is true;
- the probability that the result occurred "purely by chance";
- the probability that the alternative hypothesis is true;
- the probability that the finding will replicate.

The American Statistical Association specifically cautions against these interpretations and against making scientific decisions using a significance threshold alone. amstat.org +1

Rejecting versus failing to reject

Test outcome	Appropriate interpretation	Inappropriate interpretation
Reject H_0	The data provide evidence against the null under the test's assumptions and decision rule	The alternative is certainly true
Fail to reject H_0	The data did not provide sufficient evidence against the null under this test	The null has been proved
Statistically significant result	The test crossed a specified statistical threshold	The effect is necessarily large or practically important
Non-significant result	The test did not cross the threshold	The effect is zero

This distinction matters especially in medical research and model evaluation, where the cost of false positives and false negatives may be very different.

4.4 Type I and Type II errors

A hypothesis test makes a decision under uncertainty. Because the underlying state of the world is not known with certainty, two kinds of error are possible.

The two error types

Reality	Decision: reject H_0	Decision: fail to reject H_0
H_0 is true	**Type I error** False positive	Correct decision
H_0 is false	Correct rejection	**Type II error** False negative

Type I error

A Type I error occurs when the null hypothesis is true but the test rejects it.

Under a correctly specified test with significance level α, the Type I error probability is controlled at α, or bounded by it, depending on the test.

For example, concluding that a medical treatment has an effect when it actually has none under the null model is a false-positive conclusion.

Type II error

A Type II error occurs when the null hypothesis is false but the test fails to reject it.

Its probability is commonly denoted by β.

Statistical power is

$$\text{Power} = 1 - \beta$$

Power is the probability of rejecting the null for a specified alternative, under the assumptions of the design and analysis.

Power depends on factors such as:

- sample size;
- true effect size;
- variability or measurement noise;
- significance level;
- test choice and design.

A study can miss a genuine effect because the effect is small or the data are too noisy. That is why failure to reject the null does not establish the absence of an effect.

The trade-off

All else equal, lowering α can make a test less prone to Type I errors but can also reduce power against a given alternative. Increasing the sample size can improve power, but only if the data collection and model assumptions are appropriate.

A researcher should not choose α, sample size or analysis procedures after seeing which choices make the preferred result significant.

4.5 Confirmation bias: the central practical danger

The lecturer's discussion of confirmation bias is particularly relevant to real research.

Confirmation bias occurs when people seek, interpret, or give weight to evidence in ways that favour an existing belief or preferred explanation.

It can enter at several points:

Research stage	Example of bias	Safeguard
Hypothesis formulation	Considering only explanations that support a favoured theory	Deliberately enumerate plausible alternatives

Sampling	Selecting examples that make the method look good	Define the target population and sampling method in advance
Experimental design	Choosing a baseline likely to lose	Use strong, appropriate baselines and justify comparisons
Measurement	Measuring outcomes differently across groups	Standardise procedures and blind assessors where possible
Data cleaning	Removing inconvenient observations	Define exclusion criteria and preserve an audit trail
Analysis	Trying many configurations but reporting only the winner	Prespecify primary analyses and report relevant exploratory work
Interpretation	Treating a weak result as proof of the favoured explanation	Discuss alternative explanations and uncertainty

The lecture also warns against repeatedly sampling until a result agrees with the researcher's expectations. That is a form of selective stopping or optional sampling that can distort statistical inference when the stopping procedure is not accounted for.

Practical rule: If you would change your analysis procedure merely because the result contradicts your belief, you need a principled reason for the change—not just dissatisfaction with the result.

4.6 Experimental and control groups

Lecture 42 uses a hypothetical drug study to show why administering a treatment to a group of patients and observing improvement is not sufficient evidence of efficacy.

People can improve for reasons unrelated to the treatment:

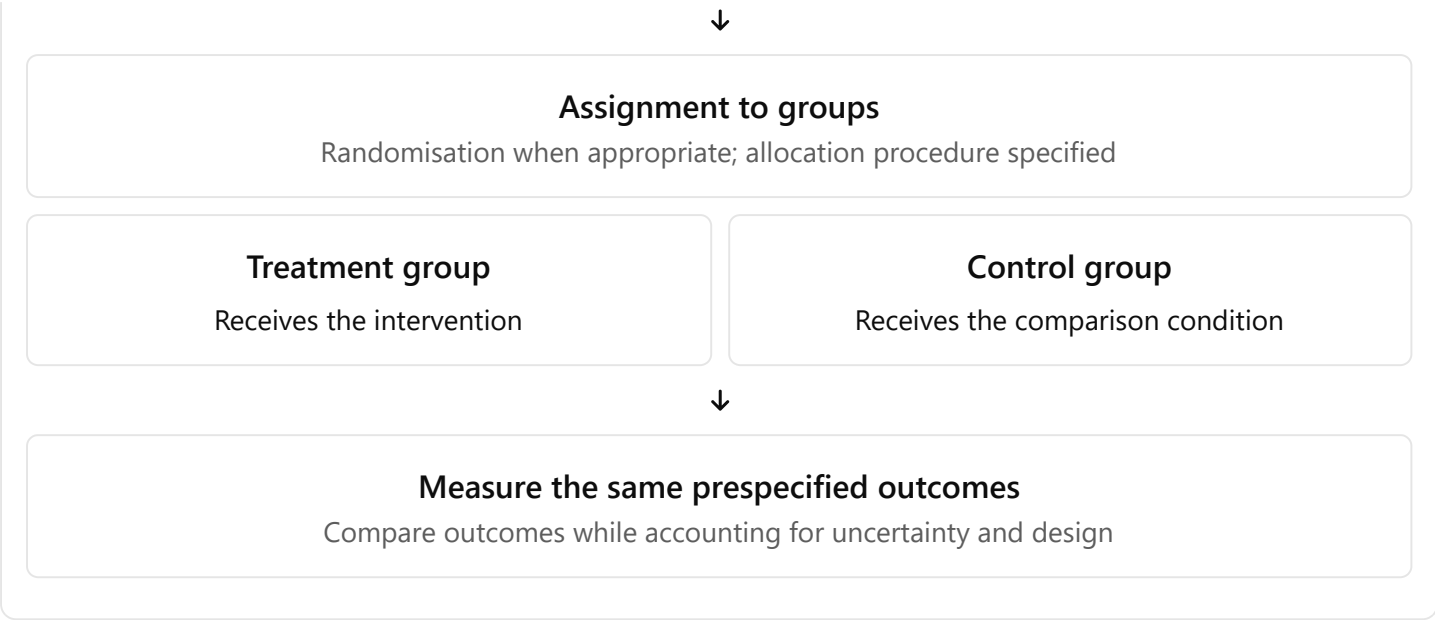
- natural recovery;
- differences in disease severity;
- other treatments or co-interventions;
- regression to the mean;
- placebo and expectation effects;
- random variation in outcomes.

A control group provides a comparison against which the intervention's effect can be assessed.

The basic design

Eligible participants or experimental units

Defined population and inclusion criteria



The treatment group and control group should differ in the intervention being evaluated as far as the design allows, while other influences are managed through randomisation, standardisation, measurement and analysis.

Why randomisation matters

Random assignment helps balance known and unknown confounding factors *in expectation*. It does not guarantee that every characteristic will be exactly balanced in a particular sample.

Random sampling and random assignment are also different:

- **Random sampling** concerns how units are selected from a population.
- **Random assignment** concerns how selected units are allocated to experimental conditions.

Random sampling can improve representativeness; random assignment can support causal interpretation of the intervention comparison. One does not automatically substitute for the other.

The lecturer's suggestion of drawing equal numbers from different groups is related to stratified sampling or blocked design. Equal allocation is not always the best design; the appropriate strategy depends on the target population, research question, costs, and expected variation.

4.7 Single-blind and double-blind designs

Blinding reduces the chance that knowledge of group assignment affects behaviour, measurement or interpretation.

- **Single-blind:** A relevant party, commonly the participant or outcome assessor, does not know the assigned condition. The exact meaning must be specified.
- **Double-blind:** Commonly, both participants and relevant researchers or assessors are blinded, but the label is not fully standardised across fields. A paper should state precisely who was

blinded.

- **Placebo:** A comparison intervention designed to resemble the treatment without containing its active component. Whether a placebo is appropriate depends on the research question and ethical context.

Blinding is especially useful when outcomes involve subjective assessment or when expectations may affect participant behaviour.

However, blinding does not replace randomisation. It also does not automatically prevent every source of bias.

Modern trial-reporting guidance asks researchers to specify who was blinded, how blinding was achieved, and how treatment allocation was implemented. Allocation concealment and blinding are distinct: concealment protects assignment before allocation, whereas blinding limits knowledge of assignments after allocation. The BMJ +1

How the principles generalise beyond medicine

The lecture's main lesson is broad: every experiment should be designed to minimise avoidable bias, but the specific safeguards depend on the field.

- In a clinical trial, a placebo may be appropriate.
- In a computer-vision experiment, the evaluator may be blinded to model identity.
- In an annotation study, annotators may be blinded to the model's prediction or the intended hypothesis.
- In a physics experiment, calibration, control conditions and blinded data analysis may be more relevant than a placebo.

The correct question is not “Did the experiment use every technique?” but “Which biases could influence this result, and which safeguards address them?”

5. Lecturer claim audit

This section distinguishes the lecturer's central teaching points from statements that need a more precise formulation.

Lecturer's claim	Verdict	Assessment and qualification
More than 25 observations makes the sample-mean distribution normal.	🟠 Potentially misleading	The CLT is an asymptotic result under suitable conditions. There is no universal sample-size cutoff; skewness, dependence and tails matter.
For a proportion, np and $n(1-p)$ should exceed 10.	🟢 Correct but simplified	This is a common approximation check, not a guarantee. More careful intervals may be

		needed for small samples or extreme proportions.
A sample of 500 with $\hat{p} = 0.66$ gives about 65% probability of falling within 0.02 of the population proportion.	🟢 Correct but simplified	This follows approximately from a normal model with p estimated as 0.66. The true sampling probability depends on the actual p and the sampling assumptions.
The 1.5 × IQR rule defines the acceptable range of data.	🟡 Context-dependent	It defines conventional Tukey fences for flagging potential outliers, not a universal boundary between valid and invalid data.
An outlier must not be hidden.	✅ Correct	Unexpected observations should be preserved and transparently reported. Whether they belong in a particular analysis requires justification.
The uncertainty in a sum or difference is the root-sum-of-squares of independent random uncertainties.	🟢 Correct but simplified	Correct for independent uncertainties interpreted as standard deviations/standard uncertainties. Correlation adds a covariance term; worst-case bounds are calculated differently.
For $z = f(x)$, uncertainty is approximately $\lvert f'(\bar{x})\rvert\sigma_x$.	🟢 Correct but simplified	This is first-order propagation. Strong nonlinearity or large uncertainty can make it inaccurate.
If a hypothesis predicts an outcome and the outcome occurs, the hypothesis is proved.	🔴 Incorrect if interpreted literally	This is affirming the consequent. Other hypotheses can predict the same outcome. The lecture explicitly rejects this invalid inference.
If a predicted outcome fails, the hypothesis is definitely false.	🟠 Potentially misleading	Modus tollens is valid for a genuine implication, but experimental predictions rely on auxiliary assumptions, implementation and measurement.
Type I error means rejecting a true null; Type II error means failing to reject a false null.	✅ Correct	These are the standard definitions. Type II error depends on the particular alternative and design.
A researcher should begin by treating the null as true.	🟡 Context-dependent	Null-hypothesis testing evaluates data under a specified null model. This is a useful testing procedure, not a general rule that researchers must believe the null is true.
Every experiment must contain a control group.	🟠 Potentially misleading	Controls are essential for many comparative experiments, but their form varies. Some research questions use observational, single-case, mechanistic or other designs where a conventional parallel control group is not feasible.

Double-blind trials are the rule everywhere in drug discovery.	🔴 Potentially misleading	Blinding is highly valuable where feasible, but not every intervention or design can be blinded. The roles blinded and the methods used must be stated explicitly.
A few people getting better after a drug does not establish efficacy.	✅ Correct	Without an appropriate comparison, natural recovery, other influences and random variation can explain improvement.
Confirmation bias can distort sampling, analysis and reporting.	✅ Correct	It is a major threat to trustworthy inference; prospective planning, transparency, independent evaluation and suitable blinding can help reduce it.

The most important external checks here are NIST's law of propagation of uncertainty, the American Statistical Association's guidance on p-values, and CONSORT guidance on randomisation, allocation concealment and blinding. NIST +2

6. Beyond the lecture — additional methodological knowledge

6.1 Effect size matters alongside statistical significance

A small p-value does not tell you whether an effect is important.

Suppose a model's accuracy rises from 90.0% to 90.2%. With a sufficiently large and suitable evaluation set, this difference might be statistically detectable. But whether it is practically useful depends on computational cost, failure severity, uncertainty and the application.

Researchers should report:

- effect estimates;
- uncertainty intervals;
- sample sizes and denominators;
- relevant comparisons and assumptions;
- practical significance, not just a significance label.

6.2 Multiple comparisons and selective reporting

Suppose you test many model architectures, datasets, random seeds and metrics, then report only the strongest result. Even if every test is conducted correctly in isolation, selecting the winner can exaggerate the apparent evidence.

Safeguards include:

- predefining primary outcomes and comparisons;
- using validation data for model selection and reserving test data for final evaluation;
- reporting the number and nature of configurations tried;

- accounting for multiplicity when appropriate;
- treating exploratory findings as exploratory until independently validated.

A held-out test set is not a licence to repeatedly inspect it and tune the method. Once the test results guide development, the set has become part of the model-selection process.

6.3 Randomisation, allocation concealment and blinding are not interchangeable

These safeguards target different problems:

Safeguard	Primary purpose
Random sampling	Improve the basis for generalising from a sample to a target population
Random assignment	Reduce systematic differences between intervention groups in expectation
Allocation concealment	Prevent knowledge of upcoming assignments from influencing enrolment or assignment
Blinding	Reduce bias arising from knowledge of the assigned condition
Standardised measurement	Reduce inconsistent outcome measurement
Prespecified analysis	Reduce opportunistic choices driven by observed results

A study can use one safeguard without using all the others. Good design begins by identifying the actual threat to inference.

6.4 Outlier analysis and robust statistics

The mean and standard deviation are sensitive to extreme observations. The median and IQR are more resistant to them, which is why they are useful in exploratory data analysis.

But robust statistics do not automatically solve every problem. A heavy-tailed distribution, a mixture of different populations, and a genuine measurement mistake are distinct situations. They should not be treated as equivalent merely because all can produce unusual values.

6.5 Uncertainty propagation is a model, not an automatic guarantee

First-order propagation works by approximating a function locally with a linear function. It is usually most reliable when input uncertainty is small relative to the scale on which the function changes.

If a function is highly nonlinear, has discontinuities, or the input distribution is far from the assumptions used, consider:

- higher-order approximations;
- Monte Carlo propagation;

- bootstrap methods when appropriate;
- explicit modelling of correlations;
- separate treatment of systematic and random components.

NIST describes the standard propagation formula as a first-order Taylor-series approximation and includes covariance terms for correlated inputs. NIST +1

7. Connection to AI/ML research

These concepts translate directly into research engineering and applied AI work.

7.1 Proportion measurement: evaluating a classifier

Imagine evaluating a medical image classifier on 200 cases. It correctly identifies 180.

$$\widehat{\text{Accuracy}} = \frac{180}{200} = 0.90$$

The observed accuracy is 90%. It is not automatically the model's accuracy on every hospital, scanner, demographic group or disease prevalence.

Questions a careful researcher asks:

- How were the 200 cases selected?
- Are the cases independent, or are multiple images from the same patient?
- Does the evaluation population resemble the intended deployment population?
- What are the confidence intervals for sensitivity, specificity and other clinically relevant proportions?
- Does the test set contain enough positive cases to estimate rare-event performance?
- Was the test set used repeatedly during model development?

For a hallucination or false-positive study, the denominator is especially important. A false-positive *rate* is not the same as the total number of false positives, and neither is interchangeable with precision.

7.2 Box plots: inspect the distribution of model performance

Suppose you train a method across 20 random seeds and compare it with a baseline.

A mean score alone may conceal:

- instability across seeds;
- a few catastrophic runs;
- substantial overlap between methods;

- performance that depends on an unusual split;
- a difference driven by one outlying run.

A box plot can make this variation visible. But remember that a box plot of seed results captures variability across those runs—not necessarily uncertainty over patients, hospitals, datasets or deployment conditions.

Practical habit: State which source of variation your plot represents.

7.3 Error propagation: derived metrics

Many AI metrics are functions of measured quantities.

For example, precision is

$$\text{Precision} = \frac{TP}{TP + FP}$$

where TP is the number of true positives and FP the number of false positives.

Both counts are uncertain estimates of performance on a target population. Simply plugging them into the formula produces a point estimate; it does not fully characterise uncertainty.

For a more rigorous analysis, use a method appropriate to the sampling design to estimate uncertainty in the derived metric. If multiple predictions come from the same patient, treating every image as independent can make the interval too narrow.

Likewise, in a medical VLM hallucination study, a hallucination rate may be calculated from annotations or labels that have their own measurement uncertainty. The metric's apparent precision cannot exceed the quality of its underlying labels and sampling design.

7.4 Hypothesis testing: an ablation is not a proof of mechanism

Suppose your hypothesis is:

> A medical VLM's false positives decrease because a new component improves evidence grounding.

You add the component and observe fewer false positives.

That is encouraging, but it does not establish the proposed mechanism. The improvement could be due to changes in prompting, model capacity, training data, thresholds, or the number of generated tokens.

A stronger design would:

1. Define the target failure mode precisely.

2. Choose a representative evaluation set.
3. Establish strong baselines.
4. Change one factor at a time where feasible.
5. Keep other conditions comparable.
6. Evaluate false positives alongside false negatives and relevant overall performance.
7. Report uncertainty and meaningful failure examples.
8. Test whether the improvement persists on independent data.
9. Avoid interpreting an observed association as proof of a mechanism.

7.5 Confirmation bias: the danger of cherry-picking a benchmark

Imagine testing several prompts, thresholds and retrieval strategies. One configuration sharply reduces hallucinations on a small set of examples.

It is tempting to present only those examples and conclude that the method solves hallucination.

A careful researcher asks:

- Were these examples selected before or after tuning?
- How many configurations were tried?
- Did the same cases guide the method's development?
- What happens on cases where the method fails?
- Does it reduce hallucinations by refusing to answer more often?
- Are false negatives or omissions increasing?
- Does the result hold on an independent evaluation set?

This is the AI research version of the lecture's warning against collecting evidence only to confirm a preferred hypothesis.

8. Research examples

Example A — Hypothetical medical VLM hallucination study

Research problem: A vision-language model generates unsupported findings from medical images.

- **Research question:** Does a proposed evidence-grounding component reduce unsupported positive findings?
- **Hypothesis:** The component will reduce the false-positive rate relative to a prespecified baseline.
- **Variables:** Model configuration, patient/case, reference findings, predicted findings, false-positive rate.
- **Method:** Evaluate baseline and proposed systems on a held-out dataset with consistent reference labels.

- **Evidence:** Compare the paired outcomes for the same cases, estimate the change and its uncertainty, and inspect the error categories.
- **Alternative explanations:** The system might produce fewer positive findings overall, increasing false negatives; the apparent improvement might depend on one dataset or annotation convention.
- **What could go wrong:** Tuning on the test set, cherry-picking examples, correlated cases, inconsistent labels, or using a metric that rewards excessive abstention.
- **Rigorous response:** Report both false positives and false negatives, clarify the denominator, preserve the test set, and validate on independent data.

Example B — Hypothetical measurement experiment

Research problem: A laboratory estimates a physical quantity indirectly from two measurements.

- **Research question:** What is the estimated quantity and its uncertainty?
- **Hypothesis:** The functional relationship $z = f(x, y)$ adequately describes the measurement process.
- **Variables:** Measured x, measured y, their uncertainty, and calculated z.
- **Method:** Measure the inputs, estimate uncertainty, apply the functional relationship, and propagate uncertainty using derivatives.
- **Evidence:** Repeat measurements, calibration records and agreement with reference values.
- **Alternative explanations:** Systematic calibration error, correlated inputs or an incorrect functional model.
- **What could go wrong:** Assuming independence when the inputs share an instrument, or reporting many decimal places without justified precision.
- **Rigorous response:** Model relevant covariance, inspect systematic effects, and report uncertainty using a clearly stated convention.

Example C — Hypothetical intervention study

Research problem: A new intervention appears to improve an outcome.

- **Research question:** Does the intervention improve the outcome relative to an appropriate control?
- **Hypothesis:** The intervention produces a meaningful improvement.
- **Variables:** Assigned condition, prespecified outcome, baseline characteristics and potential confounders.
- **Method:** Use a justified comparison design, randomisation where appropriate, consistent measurement, and blinding where feasible.
- **Evidence:** The estimated difference between groups and its uncertainty.
- **Alternative explanations:** Natural recovery, differences between groups, expectation effects or differential measurement.

- **What could go wrong:** Small samples, selective recruitment, unblinded assessment, or changing the outcome after observing the data.
- **Rigorous response:** Specify the design in advance, document assignment and exclusions, and report the results transparently.

9. Common researcher mistakes

Mistake	Why it is problematic	Better practice
Treating 25 as a universal minimum sample size	Different estimators and distributions have different requirements	Plan sample size for the actual estimand, precision and design
Reporting a proportion without its denominator	90% from 10 cases is not as informative as 90% from 10,000 cases	Report counts, denominators and uncertainty
Deleting outliers because they weaken a result	Creates selection bias and can conceal real phenomena	Investigate, justify exclusions and report sensitivity analyses
Assuming independent measurement errors	Can understate or overstate propagated uncertainty	Check shared sources of error and covariance
Treating a non-significant result as proof of no effect	Confuses insufficient evidence with evidence of absence	Consider effect estimates, intervals and statistical power
Treating a small p-value as proof	Ignores assumptions and alternative explanations	Interpret the result in the context of design and other evidence
Using only a treatment group	Cannot distinguish treatment effects from natural recovery or other influences	Use a suitable control or comparison design
Believing blinding guarantees unbiased results	Other sources of bias can remain	Combine blinding with appropriate sampling, randomisation and measurement
Changing the analysis after seeing the result	Makes selective findings appear stronger than they are	Prespecify primary analyses and disclose exploratory work
Reporting only successful experiments or configurations	Distorts the evidence base	Report relevant negative findings and all material methodological choices

10. Terminology and glossary

- **Population proportion** p — The underlying proportion in the target population; a parameter generally unknown.
- **Sample proportion** $\hat{p}$ — The observed proportion in a particular sample.
- **Standard error** — The standard deviation of a statistic's sampling distribution, estimated from data when necessary.

- **Normal approximation** — An approximation that represents a distribution by a normal distribution under suitable conditions.
- **Quartile** — A cut point in ordered data dividing the distribution into quarters, subject to the chosen quantile convention.
- **IQR** — The difference between the third and first quartiles; spread of the middle 50%.
- **Outlier** — An observation flagged as unusually distant under a stated criterion; not automatically an erroneous observation.
- **Error propagation** — The process of calculating how uncertainty in input quantities affects a derived output.
- **Sensitivity coefficient** — A derivative describing how a calculated output responds locally to a change in an input.
- **Covariance** — A measure of how two quantities vary together.
- **Null hypothesis** — A specified reference claim or statistical model assessed by a test.
- **Alternative hypothesis** — The competing claim or set of parameter values being investigated.
- p**-value** — Probability, under the null model and assumptions, of a test statistic at least as extreme as observed.
- **Type I error** — Rejecting a true null hypothesis.
- **Type II error** — Failing to reject a false null hypothesis.
- **Power** — Probability of rejecting the null under a specified alternative.
- **Confirmation bias** — A tendency to seek or interpret evidence in ways that favour prior beliefs.
- **Control group** — A comparison group used to help estimate an intervention's effect.
- **Randomisation** — Allocation by a random procedure.
- **Allocation concealment** — Protecting knowledge of future assignments during enrolment and allocation.
- **Blinding** — Preventing specified people from knowing the assigned condition.
- **Placebo effect** — Changes associated with treatment expectations or the treatment context, rather than the active treatment component alone.

11. Critical thinking questions and answers

Try answering these yourself before reading the reasoning.

Reveal each answer after thinking through the question.

1. If a sample proportion is 0.90, can we conclude that the population proportion is 0.90?

Reveal reasoning

2. Why can a sample of 500 still be inadequate?

Reveal reasoning

3. Does the 1.5 × IQR rule prove that a point is invalid?

Reveal reasoning

4. Why can subtracting two measurements create a large relative uncertainty?

Reveal reasoning

5. Why does propagation of uncertainty include covariance for correlated inputs?

Reveal reasoning

6. If a prediction derived from a hypothesis is observed, has the hypothesis been proved?

Reveal reasoning

7. Does failing to reject the null establish that an intervention has no effect?

Reveal reasoning

8. Why is a control group valuable in an intervention study?

Reveal reasoning

9. How do randomisation and blinding differ?

Reveal reasoning

10. Why is it dangerous to rerun experiments until the preferred result appears?

Reveal reasoning

11. How could a VLM appear to reduce false positives while actually becoming less useful?

Reveal reasoning

12. What should happen when a legitimate outlier contradicts the preferred hypothesis?

Reveal reasoning

Reveal all | Hide all

12. What a good researcher should notice

Five deeper lessons emerge from these lectures.

1. Uncertainty is part of the result. A point estimate without a meaningful account of uncertainty can give a false impression of precision.

2. The data do not interpret themselves. A box plot, confidence interval or p-value is a tool. Its meaning depends on how the data were generated and what assumptions are justified.

3. The experiment should constrain the researcher, not merely serve the researcher's expectations. Controls, blinding, prespecified analysis and transparent reporting help reduce opportunities for unconscious bias.

4. Negative results can be useful evidence. A carefully designed test that rules out a plausible explanation can advance knowledge even if the preferred hypothesis fails.

5. The strength of a conclusion depends on the entire chain of reasoning. Accurate measurement cannot rescue a biased sample; randomisation cannot rescue a poorly defined outcome; and a sophisticated statistical test cannot rescue an invalid research question.

13. Knowledge gaps and future learning

Topic to strengthen	Priority	Why
Statistical power, Type I/II errors and sample-size planning	**Essential now**	Needed to design informative experiments rather than merely calculate significance
Confidence intervals for proportions: Wald, Wilson and exact methods	**Essential now**	The simple normal interval can perform poorly for small samples or extreme proportions
Covariance and the full propagation-of-uncertainty formula	**Essential now**	Important whenever calculated outputs depend on correlated measurements
Randomisation, blocking and allocation concealment	**Useful soon**	Helps distinguish experimental design choices and the biases each addresses
Multiple testing, selective reporting and optional stopping	**Useful soon**	Especially important when many model configurations are evaluated
Bootstrap and Monte Carlo uncertainty propagation	**Advanced/optional**	Useful for nonlinear functions, complicated estimators and distributions where simple approximations are unreliable

Bayesian inference and likelihood-based alternatives	**Advanced/optional**	Offers other frameworks for reasoning about evidence, parameters and uncertainty

14. Final revision sheet

MUST KNOW

- A sample proportion is an estimator of a population proportion.
- $\mathrm{Var}(\hat{p}) = p(1-p)/n$ under the independent Bernoulli model.
- The IQR is $Q_3 - Q_1$; the conventional Tukey fences are $Q_1 - 1.5IQR$ and $Q_3 + 1.5IQR$.
- Independent standard uncertainties in sums and differences combine through root-sum-of-squares.
- For $z = f(x)$, first-order propagation uses the derivative $f'(x)$.
- For correlated inputs, uncertainty propagation includes covariance.
- Affirming the consequent is an invalid inference.
- Type I error = false positive; Type II error = false negative.
- Power is $1 - \beta$.
- Controls, randomisation, allocation concealment and blinding have distinct roles.

MUST UNDERSTAND

- A good approximation does not guarantee a sufficiently precise estimate.
- An outlier is not automatically an invalid observation.
- A p-value is not the probability that the null hypothesis is true.
- Failure to reject a null hypothesis does not prove the null.
- Observing a prediction does not uniquely establish the hypothesis that generated it.
- Confirmation bias can affect every stage of the research process.
- A strong experimental design considers alternative explanations before seeing the results.

MUST REMEMBER

$$\hat{p} = \frac{X}{n}$$

$$SE(\hat{p}) \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$IQR = Q_3 - Q_1$$

$$\sigma_{x\pm y} = \sqrt{\sigma_x^2 + \sigma_y^2} \quad \text{for independent inputs}$$

$$\sigma_z \approx |f'(\bar{x})|\sigma_x$$

$$\sigma_z^2 \approx \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\,\mathrm{Cov}(x, y)$$

$$\text{Power} = 1 - \beta$$

COMMON TRAPS

- "$n > 25$ guarantees normality."
- "An outlier should always be removed."
- "Independent uncertainties and correlated uncertainties are combined the same way."
- "A significant result proves the hypothesis."
- "A non-significant result proves no effect."
- "Randomisation and blinding are the same thing."
- "A model with fewer false positives must be a better model."

10-minute revision

1. Explain why sample size depends on desired precision.
2. Draw a box plot and explain its quartiles, IQR and whiskers.
3. Derive uncertainty propagation for $z = x + y$, then for $z = f(x)$.
4. Explain why a failed prediction does not always uniquely refute a scientific hypothesis.
5. Define Type I error, Type II error and power.
6. Explain how controls, randomisation and blinding address different threats to inference.
7. Name three ways confirmation bias can corrupt an AI experiment.

One-page mental model

Research question and estimand

What quantity or effect do we want to know?

↓

Design and measurement
Sampling, controls, randomisation and blinding where appropriate

↓

Data and uncertainty
Proportions, distributions, outliers and propagated uncertainty

↓

Statistical inference
Effect estimates, uncertainty, hypothesis tests and error risks

↓

Interpretation and reporting
Alternative explanations, transparent exclusions and reproducibility

This is a workflow, not a guarantee of truth. Each stage depends on the quality of the previous stages, and feedback between stages may be necessary.

15. Questions worth investigating further

1. When should a researcher use a Wilson interval rather than a Wald interval for a proportion?
2. How should sample size be chosen when the objective is to detect a clinically or scientifically meaningful effect?
3. When is first-order uncertainty propagation inadequate, and when should Monte Carlo methods be used?
4. How should a researcher handle correlated observations, such as multiple medical images from the same patient?
5. How can one quantify the impact of trying many model configurations before reporting the best result?
6. What evidence is sufficient to support a mechanistic explanation rather than merely an observed performance improvement?

These questions extend the lectures rather than contradict their central lessons.

16. Knowledge Base Update

New concepts

- Uncertainty in proportions and sample-size dependence
- Box plots, quartiles, IQR and potential outliers

- Error propagation through sums, differences and nonlinear functions
- Covariance in uncertainty propagation
- Type I and Type II errors, statistical power
- Confirmation bias in scientific testing
- Controls, randomisation, allocation concealment and blinding

Reinforced concepts

- Sampling distributions and standard errors from Week 6
- Truth versus evidence and the limits of deduction from earlier logic lectures
- Hypotheses, predictions and the possibility of alternative explanations from previous weeks
- Operationalisation and experimental design from Week 4

Important distinctions to retain

- Sample proportion versus population proportion
- Normal approximation versus precision
- Outlier flag versus justified exclusion
- Random uncertainty versus systematic error
- Standard deviation versus standard error
- Statistical significance versus practical importance
- Rejecting a null versus proving an alternative
- Random sampling versus random assignment
- Allocation concealment versus blinding
- Observed performance improvement versus evidence for a mechanism

Research habits acquired

- State the target quantity and denominator explicitly.
- Report uncertainty rather than only a point estimate.
- Preserve unexpected observations and justify exclusions.
- Check assumptions before applying formulas.
- Design experiments to discriminate between competing explanations.
- Specify the analysis and outcome criteria before seeing the result.
- Report negative findings and material limitations honestly.

Future dependencies

These ideas will be especially useful when the course develops further topics in hypothesis testing, research design, data analysis and scientific ethics.

Connections to your AI research journey

For medical VLM hallucination research, Week 7 provides a foundation for designing trustworthy comparisons: defining false-positive outcomes, choosing the correct denominator, estimating uncertainty, evaluating outliers and disagreements in annotations, comparing models fairly, and resisting the temptation to report only the most favourable configuration.

The central discipline is to make your experiment capable of showing that your hypothesis is wrong—not merely capable of producing evidence that appears to support it.

Recommended study actions

1. Rework the $57/150$ proportion-test example and explain the p-value in your own words.
2. Recalculate the box-plot example from its original data; notice the transcript's inconsistency rather than memorising its quartiles.
3. Derive the propagation formula for a product and explain when the independence assumption matters.
4. Design a small hypothetical VLM ablation study with a primary outcome, baseline, control of confounders and a plan for reporting negative results.
5. Review the definitions of Type I error, Type II error, power and confirmation bias until you can explain them without notes.

‹/›